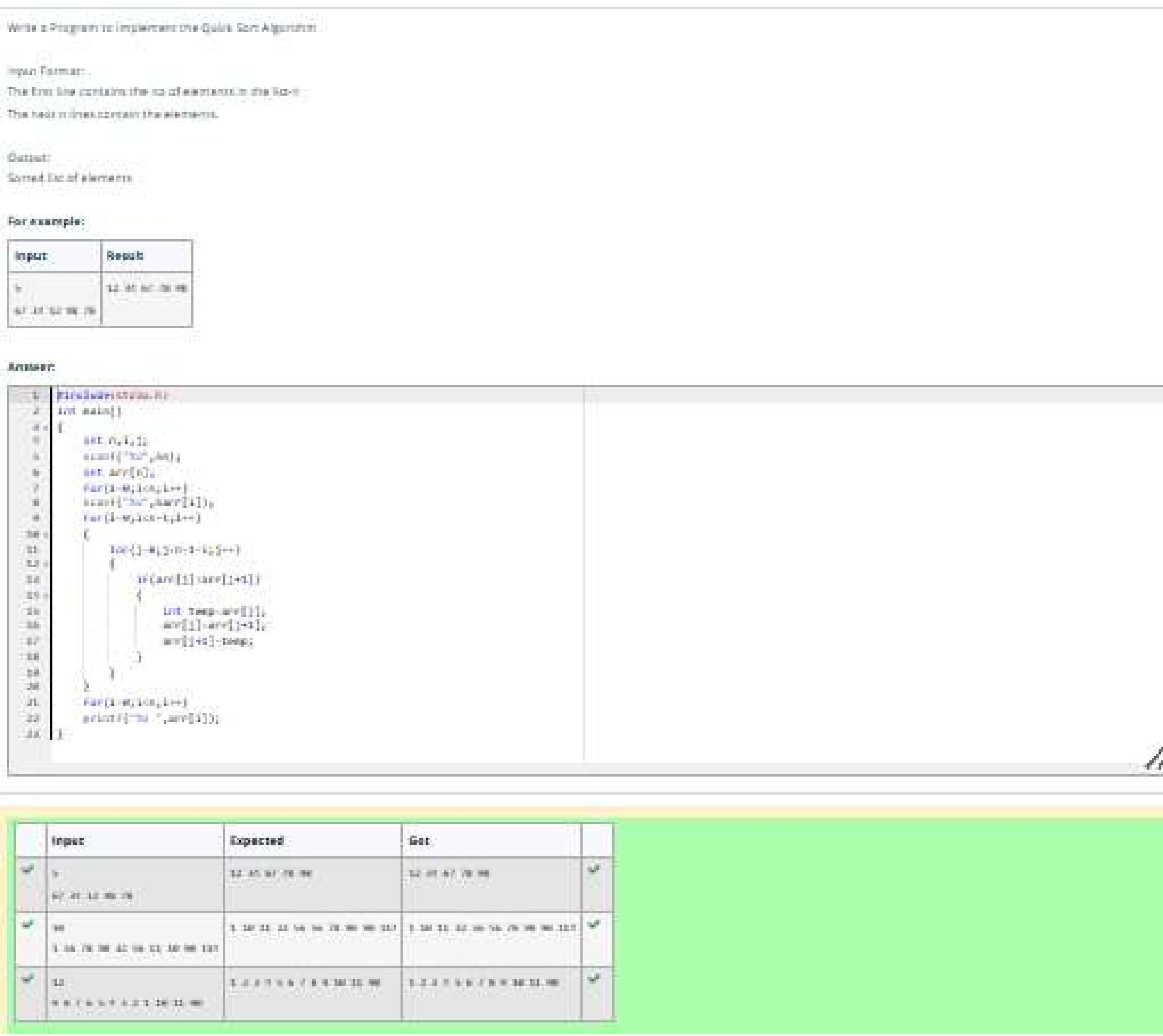


Write a Program to implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

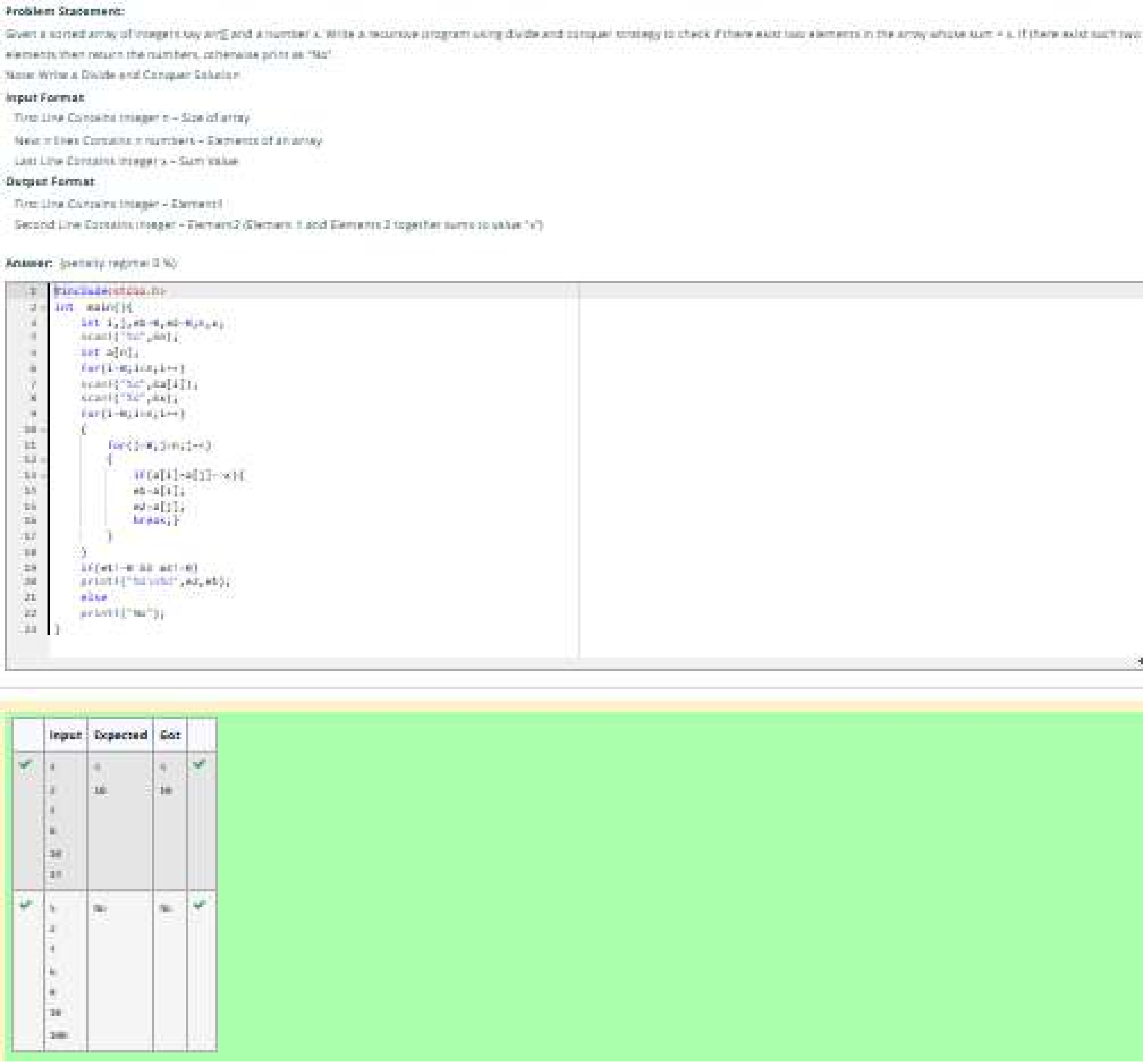
For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```
#include<stdio.h>
int main()
{
    int n,i,j;
    scanf("%d",&n);
    int arr[n];
    for(i=0;i<n;i++)
    scanf("%d",&arr[i]);
    for(i=0;i<n-1;i++)
    {
        for(j=0;j<n-i-1;j++)
        {
            if(arr[j]>arr[j+1])
            {
                int temp=arr[j];
                arr[j]=arr[j+1];
                arr[j+1]=temp;
            }
        }
    }
    for(i=0;i<n;i++)
    printf("%d ",arr[i]);
}
```

	Input	Expected	Got	
✓	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✓
✓	10 1 56 78 90 32 56 11 10 90 117	1 10 11 32 56 56 78 90 90 117	1 10 11 32 56 56 78 90 90 117	✓
✓	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✓



Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two
elements then return the numbers, otherwise print as "No"

Note: Write a Divide and Conquer Solution

Input Format:

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format:

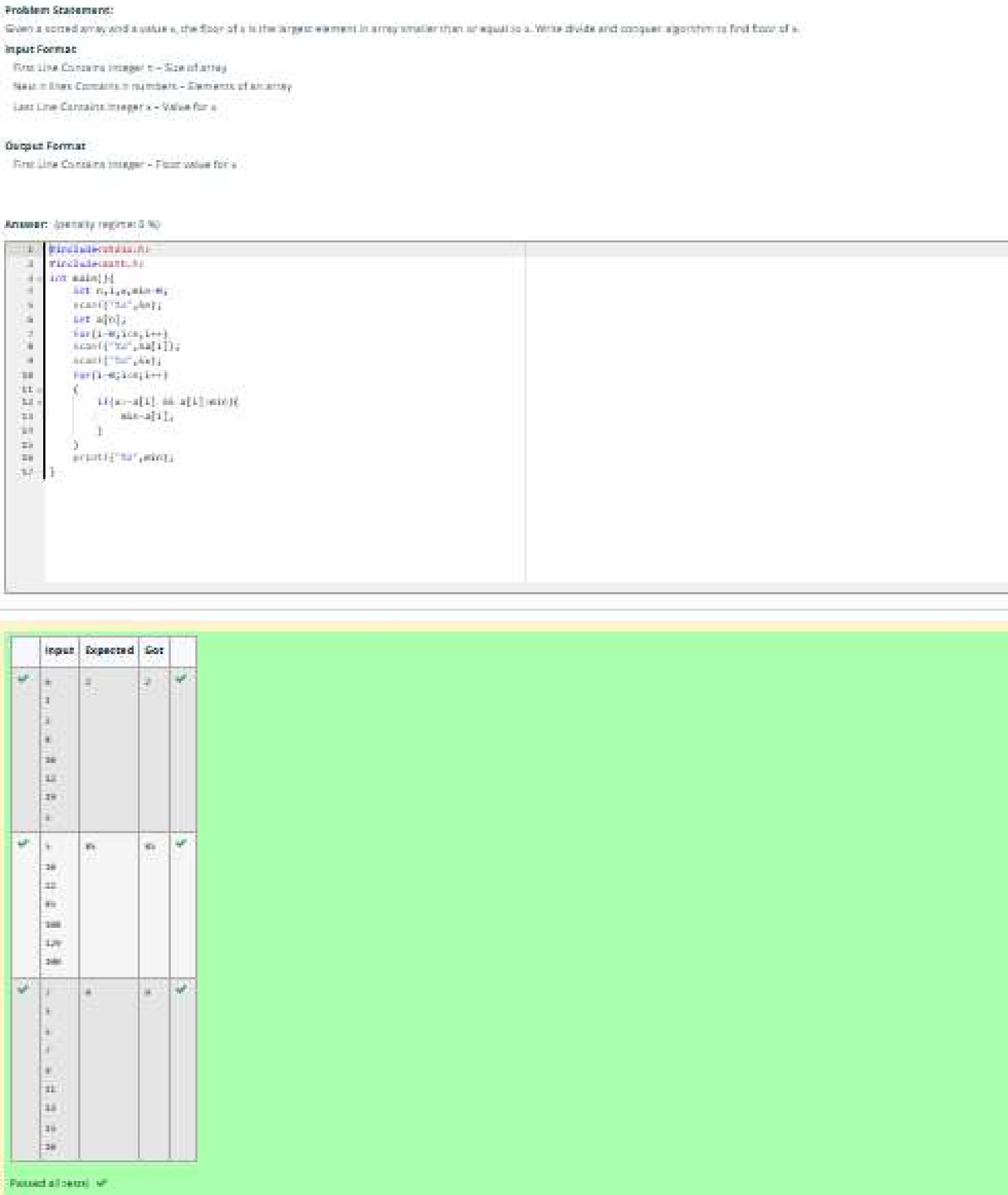
First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int  main(){
    int i,j,e1=0,e2=0,n,x;
    scanf("%d",&n);
    int a[n];
    for(i=0;i<n;i++)
    scanf("%d",&a[i]);
    scanf("%d",&x);
    for(i=0;i<n;i++)
    {
        for(j=0;j<n;j++)
        {
            if(a[i]+a[j]==x){
            e1=a[i];
            e2=a[j];
            break;}
        }
    }
    if(e1!=0 && e2!=0)
    printf("%d\n%d",e2,e1);
    else
    printf("No");
}
```

	Input	Expected	Got	
✓	4 2 4 6 10 16	6 10	6 10	✓
✓	5 2 4 6 8 10 100	No	No	✓



Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

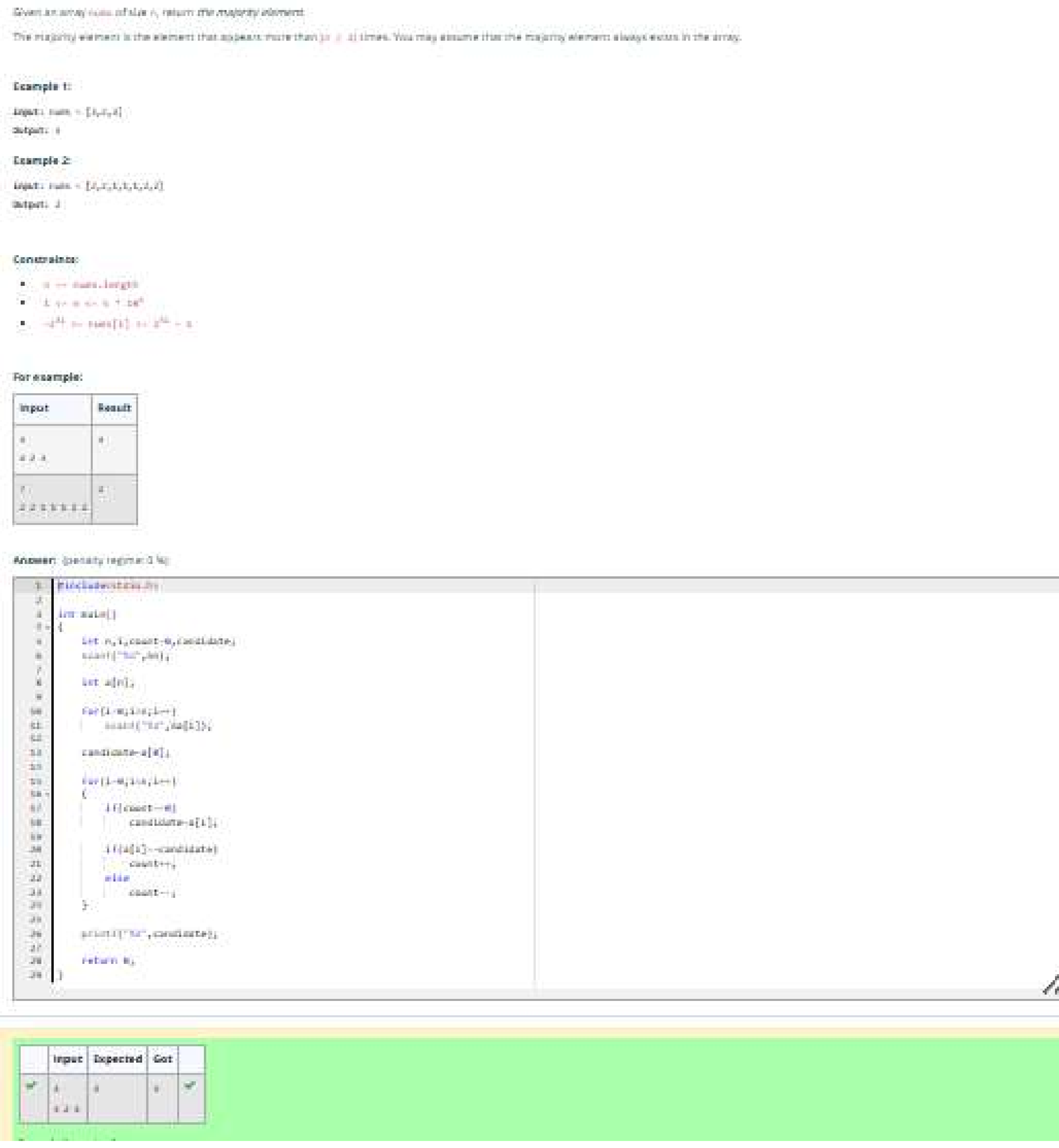
First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
#include<math.h>
int main(){
    int n,i,x,min=0;
    scanf("%d",&n);
    int a[n];
    for(i=0;i<n;i++)
    scanf("%d",&a[i]);
    scanf("%d",&x);
    for(i=0;i<n;i++)
    {
        if(x>=a[i] && a[i]>min){
            min=a[i];
        }
    }
    printf("%d",min);
}
```

	Input	Expected	Got	
✓	6 1 2 8 10 12 19 5	2	2	✓
✓	5 10 22 85 108 129 100	85	85	✓
✓	7 3 5 7 9 11 13 15 10	9	9	✓

Passed all tests! ✓



Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10`^{4}
- `-2`^{31} `<= nums[i] <= 2`^{31} `- 1`

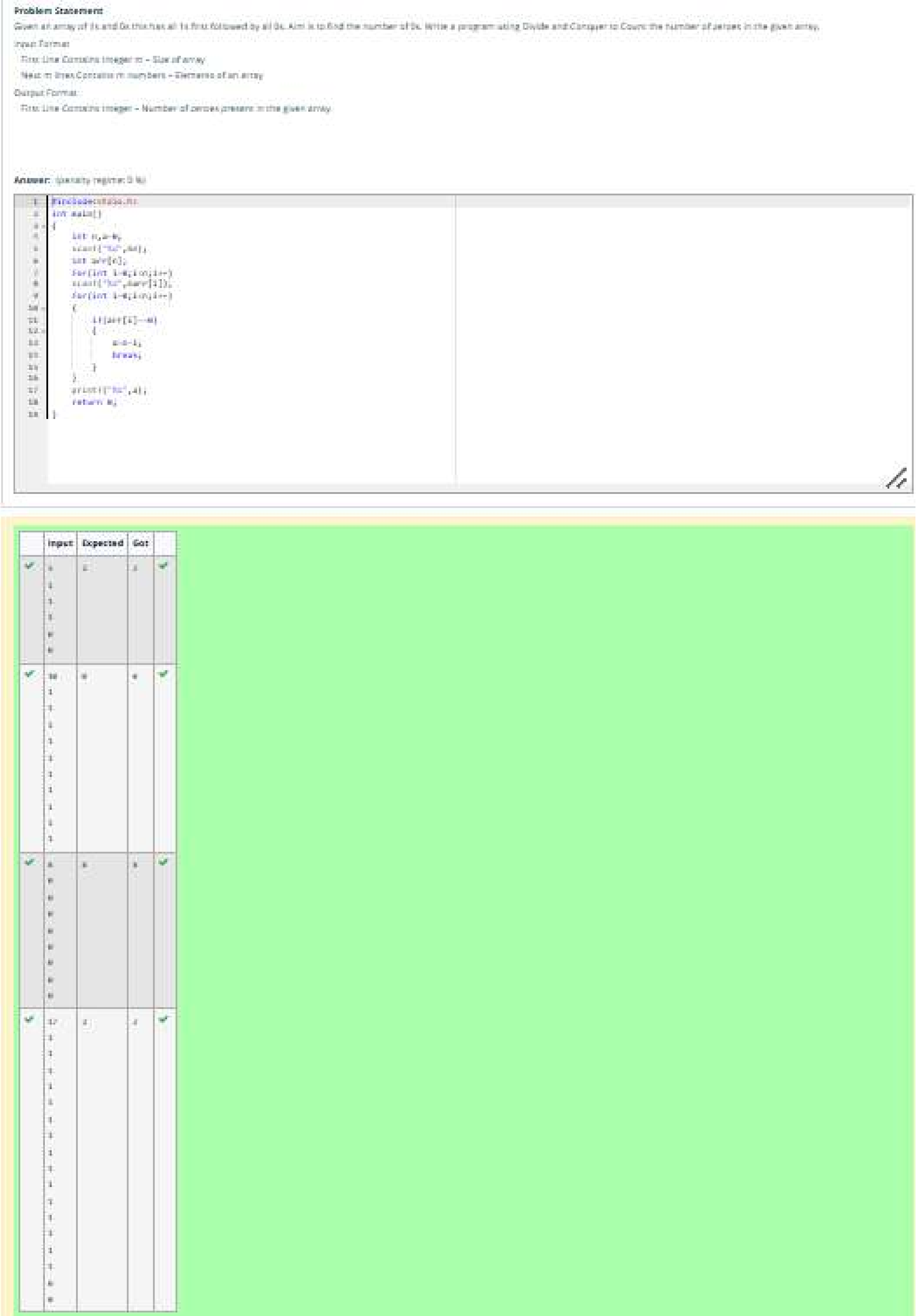
For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```
#include<stdio.h>

int main()
{
    int n,i,count=0,candidate;
    scanf("%d",&n);

    int a[n];

    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    candidate=a[0];

    for(i=0;i<n;i++)
    {
        if(count==0)
            candidate=a[i];

        if(a[i]==candidate)
            count++;
        else
            count--;
    }

    printf("%d",candidate);

    return 0;
}
```

	Input	Expected	Got	
✓	3 3 2 3	3	3	✓



Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main()
{
    int n,a=0;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++)
    scanf("%d",&arr[i]);
    for(int i=0;i<n;i++)
    {
        if(arr[i]==0)
        {
            a=n-i;
            break;
        }
    }
    printf("%d",a);
    return 0;
}
```

	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓
✓	10 1 1 1 1 1 1 1 1 1 1	0	0	✓
✓	8 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓